

EX-NO:08**XML AND XQUERY OPERATIONS****24-11-23****AIM:**

To create XML document and validate the xml document and also performing the XQUERY commands.

XML Document (Location.xml):

```
<?xml version="1.0"?>
```

```
<students>
```

```
  <student id="10001">
```

```
    <name>John Doe</name>
```

```
    <address>
```

```
      <street>456 University Ave</street>
```

```
      <city>College Town</city>
```

```
      <state>CA</state>
```

```
      <zip>98765</zip>
```

```
    </address>
```

```
  </student>
```

```
  <student id="10002">
```

```
    <name>Jane Smith</name>
```

```
    <address>
```

```
      <street>789 Elm St</street>
```

```
      <city>Cityville</city>
```

```
      <state>NY</state>
```

```
      <zip>54321</zip>
```

```
    </address>
```

```
  </student>
```

```
  <student id="10003">
```

```
    <name>Bob Johnson</name>
```

```
    <address>
```

```
      <street>321 Oak Blvd</street>
```

```

    <city>Treetown</city>

    <state>IL</state>

    <zip>67890</zip>

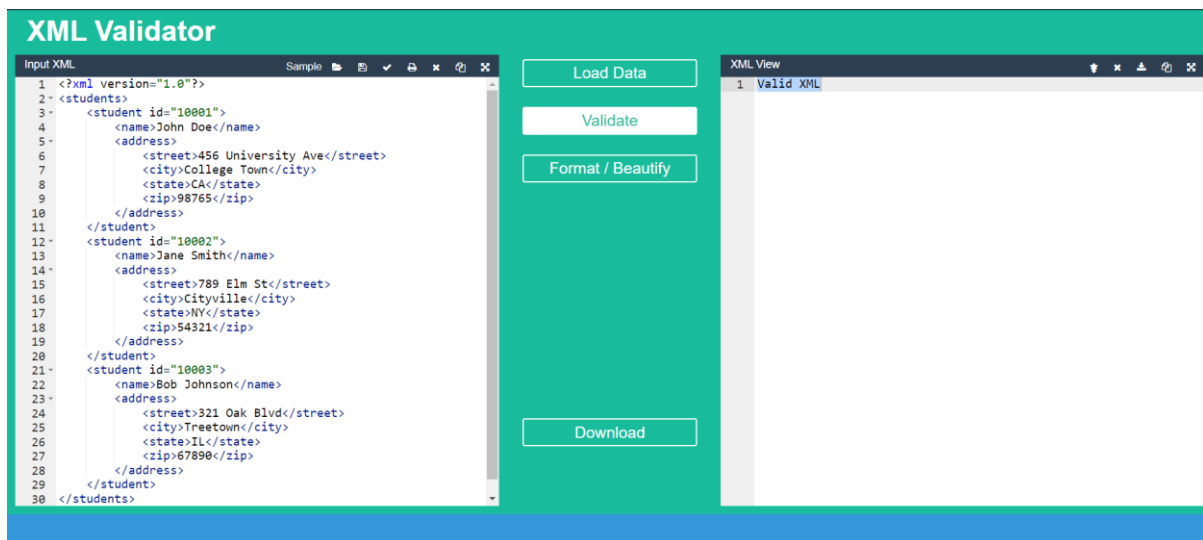
  </address>

</student>

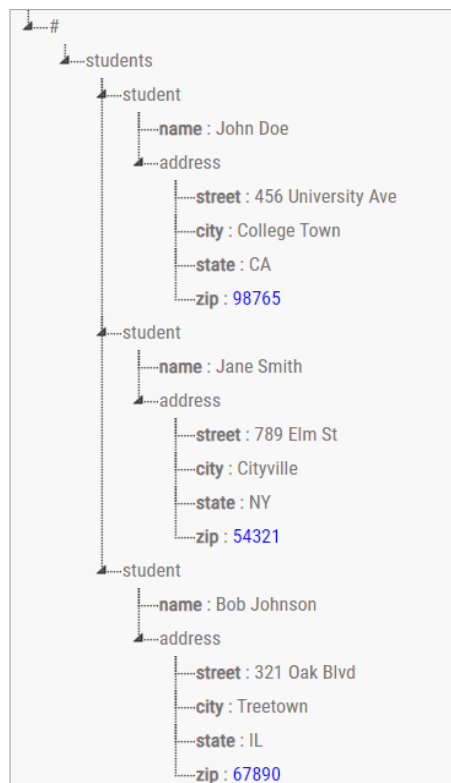
</students>

```

XML Validation: Validating the XML Document.



XML Tree: Representing the xml document in tree structure.



XML Database:**Use Basex :**

```
C:\Users\Janani>BaseX
BaseX 10.7 [Standalone]
```

Creating database and location.xml

```
> CREATE DB mydatabase C:\Program Files (x86)\BaseX\location.xml
Database 'mydatabase' created in 141.29 ms.
```

Open Database (mydatabase)

```
> OPEN mydatabase
Database 'mydatabase' was opened in 0.06 ms.
```

XQUERY QUERIES:**To retrieve all Students:**

```
> xquery //students
<students>
  <student id="10001">
    <name>John Doe</name>
    <address>
      <street>456 University Ave</street>
      <city>College Town</city>
      <state>CA</state>
      <zip>98765</zip>
    </address>
  </student>
  <student id="10002">
    <name>Jane Smith</name>
    <address>
      <street>789 Elm St</street>
      <city>Cityville</city>
      <state>NY</state>
      <zip>54321</zip>
    </address>
  </student>
  <student id="10003">
    <name>Bob Johnson</name>
    <address>
      <street>321 Oak Blvd</street>
      <city>Treetown</city>
      <state>IL</state>
      <zip>67890</zip>
    </address>
  </student>
</students>
Query "Janani" executed in 324.71 ms.
```

To retrieve the students name of first record:

```
> xquery (//student)[1]/name
<name>John Doe</name>
Query "Janani" executed in 2.75 ms.
```

To retrieve the students with specific id:

```
> xquery //student[@id = '10002']
<student id="10002">
  <name>Jane Smith</name>
  <address>
    <street>789 Elm St</street>
    <city>Cityville</city>
    <state>NY</state>
    <zip>54321</zip>
  </address>
</student>
Query "Janani" executed in 4.28 ms.
```

To retrieve the student with specific address and city:

```
> xquery //student[address/city = 'Cityville']
<student id="10002">
  <name>Jane Smith</name>
  <address>
    <street>789 Elm St</street>
    <city>Cityville</city>
    <state>NY</state>
    <zip>54321</zip>
  </address>
</student>
Query "Janani" executed in 4.13 ms.
```

To retrieve all the students's name:

```
> xquery //student/name
<name>John Doe</name>
<name>Jane Smith</name>
<name>Bob Johnson</name>
Query "Janani" executed in 2.31 ms.
```

To retrieve the distinct values of students address and city:

```
> xquery distinct-values (//student/address/city)
College Town
Cityville
Treetown
Query "Janani" executed in 63.4 ms.
```

To use for condition and return students name:

```
> xquery for $student in //student where $student/address/city = 'Cityville' return $student/name
<name>Jane Smith</name>
Query "Janani" executed in 19.75 ms.
```

To modify the name of the student:

```
> xquery replace value of node (//student)[2]/name with 'Janani'

Query "Janani" executed in 25.1 ms.
> xquery (//student)[2]/name
<name>Janani</name>
Query "Janani" executed in 2.78 ms.
```

Using orderby command:

```
> xquery for $student in // student order by $student/name return $student
<student id="10003">
  <name>Bob Johnson</name>
  <address>
    <street>321 Oak Blvd</street>
    <city>Treetown</city>
    <state>IL</state>
    <zip>67890</zip>
  </address>
</student>
<student id="10002">
  <name>Janani</name>
  <address>
    <street>789 Elm St</street>
    <city>Cityville</city>
    <state>NY</state>
    <zip>54321</zip>
  </address>
</student>
<student id="10001">
  <name>John Doe</name>
  <address>
    <street>456 University Ave</street>
    <city>College Town</city>
    <state>CA</state>
    <zip>98765</zip>
  </address>
</student>
Query "Janani" executed in 8.05 ms.
```

Displaying first address of the student:

```
> xquery //student/address[1]
<address>
  <street>456 University Ave</street>
  <city>College Town</city>
  <state>CA</state>
  <zip>98765</zip>
</address>
<address>
  <street>789 Elm St</street>
  <city>Cityville</city>
  <state>NY</state>
  <zip>54321</zip>
</address>
<address>
  <street>321 Oak Blvd</street>
  <city>Treetown</city>
  <state>IL</state>
  <zip>67890</zip>
</address>
```

Displaying all the address in the database:

```
> xquery //address/*
<street>456 University Ave</street>
<city>College Town</city>
<state>CA</state>
<zip>98765</zip>
<street>789 Elm St</street>
<city>Cityville</city>
<state>NY</state>
<zip>54321</zip>
<street>321 Oak Blvd</street>
<city>Treetown</city>
<state>IL</state>
<zip>67890</zip>
Query "Janani" executed in 5.47 ms.
```

Displaying first 2 students:

```
> xquery //student[position()<=2]
<student id="10001">
  <name>John Doe</name>
  <address>
    <street>456 University Ave</street>
    <city>College Town</city>
    <state>CA</state>
    <zip>98765</zip>
  </address>
</student>
<student id="10002">
  <name>Janani</name>
  <address>
    <street>789 Elm St</street>
    <city>Cityville</city>
    <state>NY</state>
    <zip>54321</zip>
  </address>
</student>
Query "Janani" executed in 5.52 ms.
```

Displaying the Last student:

```
> xquery //student[last()]
<student id="10003">
  <name>Bob Johnson</name>
  <address>
    <street>321 Oak Blvd</street>
    <city>Treetown</city>
    <state>IL</state>
    <zip>67890</zip>
  </address>
</student>
Query "Janani" executed in 3.82 ms.
```

Using Logical and:

```
> xquery //student[address/city = 'Cityville' and address/zip = '54321']
<student id="10002">
  <name>Janani</name>
  <address>
    <street>789 Elm St</street>
    <city>Cityville</city>
    <state>NY</state>
    <zip>54321</zip>
  </address>
</student>
Query "Janani" executed in 3.65 ms.
```

RESULT:

Thus, the xml document is created and validated and the XQuery commands.

OBSERVATION	RECORD	SIGNATURE